better the appropriate legislation with the definition of 'land grabber', 'land grabbing' and 'land grabbing cases'.

Andhra Pradesh has become the first state in India to pilot blockchain technology in two departments, and plans to deploy it across the administration.

Current state of the land management system

Land management entails the registration and management of all land deeds undertaken in any geographical location of a governing authority. Various government departments and local bodies maintain these records. The registration of land records does not authenticate the 'land title', which sometimes leads to title disputes and litigation.

The current land registration system in India involves a paper-based approach that is susceptible to fraud, errors, and security risks. The process is also time-consuming and involves huge documentation, which contributes to fraudulent activities.

One survey estimated that 70% of landowners are vulnerable to manipulation by individuals or mediators with stakes, who challenge the validity of their ownership. There are a huge number of examples across India where land has been grabbed by forging records.

Of late, there is an increase in complaints of irregularities and corruption in updating land records. Many Indians are eagerly waiting for corrections in their land records.

In today's scenario, the seller and buyer meet at the registration office, share the physical documents in the presence of the Registrar, and complete the land registration. The transaction primarily happens through the exchange and verification of physical copies.

The major issues with land registration in India are:

- Unclear land titles, leading to huge GDP loss
- Land grabbing
- Duplicate land records
- Land records are maintained in silos; hence, updating and verifying the records is a challenge
- Poor record-keeping, with stakeholders often dealing with inaccurate data
- The database method used for storing data is inadequate in terms of data security
- Multiple land web portals, and none of them provide a single source of truth

All the above facts demand the government to develop a secured, tamper-proof, paperless, and long-lasting online solution to manage land registration and land records. The best solution for this is implementation of blockchain technology-based land management.

Blockchain-based land management solution for the future

Blockchain technology helps establish trust, transparency and accuracy in maintaining land records and building a robust land management system. Implementation of blockchain in land records provides:

- Integrity
- Transparency
- Elimination of intermediaries
- Speed
- Elimination of documentation work
- Minimisation of fraud

The blockchain-enabled land management system is a decentralised public ledger spread over a distributed network that records every transaction associated with a land property in an autonomous and efficient manner. It enhances the accuracy and security of land records.

Blockchain works by validating transactions through a distributed network to create a permanent, verified, and unalterable ledger of information. Figure 2 shows the end-to-end flow of the blockchain-based land registration process.

The steps followed in this process are:

- The seller requests that the invoice be published by the purchaser. Seller stores the relevant factoring agreement in a smart contract on a public blockchain.
- Purchaser examines the smart contract, notices that the invoice has been factored and pays the invoice amount.
- After obtaining the signature of the purchaser and seller in the sale deed, the scanned document is moved into the blockchain network to create a block.
- The certificates issued by the revenue department are stored in the blockchain and can be used by other agencies like the bank for any verification process during a land transaction.
- Verification of the details is done using the blockchain data. After the approval of transactions in the respective database, such as completion of registration or approval of a loan by the bank, the transaction details are stored in the blockchain.

Once the block is created, it cannot be tampered with or edited. The chain of blocks is created every time the property title is changed from one person to another.

Also, the existing history of transactions on a piece of land first needs to be inserted into the blockchain after approval by revenue functionaries in the state. The approved data can be digitally signed and stored.

Blockchain technology helps to set up:

Tamper-proof land records: Land records are secure, tamper-proof and immutable. This reduces the scope of forgery or manipulation.

Unified registration system: A single digital platform removes bureaucratic silos and helps in driving greater efficiency and reducing inaccuracies/forgeries.